Room plan: bridges (pick up every counter)

The plan for Wednesday, November 18, 2026 (proposed; James agreed to Weeks 7 and 10 next on November 3). Week 10, Bridges: packets for K–1, 2–3 and 4–5, and the facilitator guide, whose floor-town launch this plan uses. Unpiloted, at the circle and at home.

Why this follows the take-away games session

Last week's take-away games had children take counters from piles in turns. Today the counters sit on bridges, and the town decides which counter you may take next: cross a bridge, pick up its counter, and that bridge is closed. The finale is the guide's bridge game, which is a take-away game played on a map.

This week shares its theorem with Week 2's lamps, which went poorly on paper because nothing stopped a child changing just the lamp they wanted. Here the rule is physical: a crossed bridge loses its counter into your hand.

For adults: in a town in one piece, a walk that crosses every bridge exactly once exists exactly when zero or two islands have an odd number of bridges. With zero it can start anywhere and ends where it started; with two it must start at one odd island and end at the other. Reason: a walk passing through an island uses its bridges in pairs, in and out. On the floor town, purple and yellow have 3 bridges and the rest have 2, so the walk runs from purple to yellow or back. The opener asks a different question, visiting every house once, which has no counting rule. That contrast is worth saying to the 4–5 table.

Prep checklist for Wednesday afternoon

Print single-sided on Letter paper at 100% ("Actual size"); "Fit to page" shrinks the bridges below counter size. The packets print fine in black and white; the opener needs colour. The guide's counts assume 4 children at K–1, 4 at 2–3 and 3 at 4–5; the 2–3 count leans up, since most of the room is there.

- K–1 packet, 12 pages: 5 copies
- Grades 2–3 packet, 13 pages: 7 copies
- Grades 4–5 packet, 12 pages: 4 copies
- Facilitator guide, 8 pages: one per adult (3)
- This room plan: one per adult (3)
- Opener sheet, 2 pages, in colour, cut in half: page 1 (the main puzzle, two per page) 5 copies, page 2 (three houses on top, follow the paths below) 3 copies
- Counters about 1 inch or smaller, about 150, in three bags: K–1 50, 2–3 60, 4–5 40 (guide, section 2)
- Tokens unlike the counters, such as pattern-block triangles: 10 (2 per pair, one walks and one marks the start)
- For the floor town: masking tape, a pad of sticky notes, a thick marker, a piece of string about 4 feet long, and five coloured sheets (green, blue, red, yellow, purple; November 4's chair sheets will do)
- Coloured pencils, pencils, erasers, scrap paper, small whiteboards
- A timer everyone can see
- Three early-finisher cards, handwritten is fine (see below)
- Optional, only for K–1 Problem 7 and 2–3 Problem 8: 20 yellow hexagons and about 30 craft sticks. Skip them if prep is short.
- Five-minute check the day before: the guide's counter test. Put a counter on all 8 bridges of K–1 page 9, second town. If they crowd, use smaller tokens (buttons, beans).

Prep time: about 15 minutes to print, 10 to bag the counters, 5 for the counter test, and about 10 at the room to tape the floor town (untried; one adult). Nothing to cut except the opener sheet.

Before the children arrive

- **Tape the floor town** (the house town from packet page 1, as in the guide), in open floor about 8 by 12 feet. Each coloured sheet is an island; tape it down on all four edges so nobody slips, and write its packet letter on it in big marker:
    - blue **D** and red **E** on the floor, 4½ feet apart, centre to centre (the bottom corners);

    - purple **B** 4½ feet straight up from blue, yellow **C** 4½ feet straight up from red (the square);
    - green **A** centred above, about 4 feet past the purple–yellow line (the roof).
    - Six tape bridges, island edge to island edge, each about 3½ feet: D–E, D–B, E–C, B–C, B–A, C–A. A sticky note in the middle of each.
- On every table: opener half-sheets and pencils, so the first child in starts on the puzzle; that band's counter bag, two tokens per pair, packet page 1 face down (one page at a time, both pages where a problem continues), coloured pencils.
- The timer where everyone can see it. The string and spare sticky notes by the floor town.
- Early-finisher cards, one per table:
    - K–1: "Draw a town with 4 islands. Put a counter on every bridge. Can your partner pick up every counter?"
    - 2–3 and 4–5 (the guide's bridge game): "House town, Problem 1. Counters on every bridge, one token on an island. Take turns: move the token over one bridge and keep its counter. If you cannot move, you lose. Does it matter where the token starts?"

Opener: Pat's mail route (8 minutes)

A soft logic puzzle on the tables as children arrive, then walked on the floor town, whose five islands are Pat's houses. Pat's route starts with the guide's first legal move and leaves two notes on the floor, so the puzzle leads straight into the launch question. An adult reads the clues aloud at any table that wants it.

The puzzle (most of the room). Pat the mail carrier brought one letter to each of five houses (the house town: green on top, purple and yellow in the middle, blue and red at the bottom), one house after another. Number the houses in the order Pat went.

1. Pat went to the red house right after the blue house.
2. Pat went to the green house right after the yellow house.
3. Pat went to the yellow house some time after the red house.
4. The green house was not the last house.

Answer: blue 1, red 2, yellow 3, green 4, purple 5. It is the only answer, and every clue is needed, whether or not a child also assumes Pat walks along the paths (checked by computer, mail-route_check.py). Reading clue 3 as "right after" gives the same answer.

Hint ladder (give one step, then walk away):

1. "Which two houses always go together? Which other two?" (Blue then red; yellow then green.)
2. "Clue 3: which pair comes first?" (Blue and red come before yellow and green.)
3. "Where can purple go? Look at clue 4." (Not before blue or between the pairs, since then green is last: purple is 5.)

Easier (younger children, or anyone stuck at step 1). Three houses: green on top, purple and yellow below. Pat went to the yellow house right after the purple house. Pat did not start at the green house. *Answer:* purple 1, yellow 2, green 3.

Harder (older children, or anyone done in two minutes). The same five houses, and now the paths matter: from each house Pat walked along one path straight to the next. Pat started at a bottom house. Pat went to the red house right after the blue house. The top house was not first and not last. *Answer:* the same route. Red cannot be first, so blue is; from red the only new house is yellow; green is not last, so green then purple. Without the path rule it has 4 answers.

Early finishers. "Cross out clue 4. Find every order that fits now. Write a new clue that makes it exactly one again." (Three orders: purple first, purple third, or purple last; two if the child keeps to the paths. "The purple house was the last house" fixes it, checked, and so do others.)

On the floor (minute 6). Everyone to the floor town. Adult: "These are Pat's houses. Every bridge has a note on it. When Pat crosses a bridge, Pat picks up its note. A bridge with no note is closed." One solver plays Pat and starts on blue. The room says each clue as Pat walks, thumbs up or down: blue, along the bottom bridge to red, up to yellow, over the roof to green, down to purple. Pat holds 4 notes. Bridge question: "Every house got its letter. Two notes are left. Can Pat keep walking and pick them both up?"

The hour

Time	Everyone	Jobs
0:00–0:08	**Opener: Pat's mail route.** Puzzle on the tables as children arrive; at minute 6, Pat walks the answer on the floor town and picks up 4 notes.	Solvers check each other's sheets; adults give one hint step at a time.
0:08–0:12	**Launch at the floor town** (guide, steps 4–5). Pat goes on from purple and gets stuck with a note left either way. "Can Pat cross an empty bridge?" (No.) "Would a different start pick up every note? That is today's question." Everyone stands by the island they would start from; nobody says who is right. Practise the reset signal: "Bridges up", both hands on heads. Notes back; off to tables.	Pat walks; everyone else predicts out loud.
0:12–0:20	**Round 1.** K–1 Problem 1 (page 1). 2–3 Problem 1 (page 1). 4–5 Problem 1 (page 1).	**Walker** moves the token; **Keeper** resets counters, says "stop" at a closed bridge, and at 2–3 writes the walk. 4–5 adds a **Referee**. Swap each town.
0:20–0:28	**Round 2.** K–1 Problem 2 (page 2). 2–3 Problem 2 (pages 2–3). 4–5 Problem 2 (pages 2–3).	Same jobs, swapped. 4–5 rotates the referee.
0:28–0:33	**Back to the floor.** A pair who found a start for the house town (Problem 1: purple or yellow) walks it; the room watches the notes. Lay the string beside the purple–yellow tape as a second bridge, with a note: "Now try from blue." It works (blue, red, yellow, purple, yellow, green, purple, blue). Ask "Why did one bridge change that?", then take the string away.	A new Pat from each table; the pair says where to start.
0:33–0:41	**Round 3.** K–1 Problem 3 (pages 3–4). 2–3 Problem 3 (pages 4–5). 4–5 Problem 3 (pages 4–5).	Same pairs, swapped jobs.
0:41–0:49	**Round 4.** K–1 Problem 4 (pages 5–6), or Problem 7 (page 10) with hexagons. 2–3 Problem 4 (pages 6–7), or Problem 6 (page 9) for a pair ready to say the rule. 4–5 Problem 4 (page 5), then Problem 5, Königsberg (page 6). At 0:47 each adult asks the guide's sharing question: "Show me a town where nobody can pick up every counter. How do you know?"	Same jobs, swapped.
0:49–0:55	**Floor finale: the bridge game.** Notes on all six bridges. Two teams, one Pat in the middle as the token. Teams take turns telling Pat which bridge to cross, and keep its note. A team that cannot move loses. Game 1 starts on blue, game 2 on purple.	One caller per team per turn; a referee checks each bridge has a note.
0:55–1:00	Counters back into the bags (50, 60, 40); tokens and pages collected.	Each pair counts one pile; one child per table collects pages.

If it goes sideways

- A table finishes early: hand out its early-finisher card.
- A pair is stuck: "Put the counters back and start on a different island", nothing more (guide, hint 1).
- Writing walks as letters slows the 2–3 table: the child says the letters and the adult writes.
- The room gets loud: "Bridges up", then go straight to the next floor activity.
- No floor space: the guide's fallback. Counters on K–1 page 1's house town at one table; Pat is a token.
- Finale: from blue, red or green the first team can force a win; from purple or yellow the second can (checked). Do not say so.
- Jay and Rose: the use log has no bridges entry. If they have seen it (at home or in Bridge Courier), make them floor captains at 0:28 and 0:49, and 2–3 checkers who may say "try again" but not why.

How it went

(Add a dated line after the session.)